

Revised Generation Interconnection System Impact Study Report

For Queue Project Q24-2482

Bedford 138 kV
November 19, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2482. The Interconnection Customer proposes a 786.0 MW natural gas combined cycle facility interconnecting at the Bedford 138 kV substation in the Kentucky area. The request was received on December 02, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 440 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$222,465,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
BEDFORD - BAXTER 138 kV	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	372.3	223	99.9
FIRSTENERGY ALBRIGHT - BEDFORD 138 kV L	BAXTER - BEDFORD 138 kV	352.7	235	99.9
BAXTER - MARACAIBO 138 kV line	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	349.5	231	99.9
POSO MTN - EL PORTA 138 kV	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	230.0	181	42.7
FIRSTENERGY ALBRIGHT - ZENITH 138 kV line	FIRSTENERGY ALBRIGHT - SS3145EAGLE ROCK TAP 2 138 kV	207.0	222	43.8
ARVERNE - SS3145EAGLE ROCK TAP 2 SUBSTATION 138 kV	FIRSTENERGY ALBRIGHT - ZENITH 138 kV line	186.9	227	39.0
FIRSTENERGY ALBRIGHT SUBSTATION - SS3145EAGLE ROCK TAP 2 138 kV	ZENITH - FIRSTENERGY ALBRIGHT 138 kV	186.9	246	39.0
ZENITH - ROYALTY 138 kV L	FIRSTENERGY ALBRIGHT - SS3145EAGLE ROCK TAP 2 138 kV line	180.9	238	43.8
EL PORTA - UNIVERSIITY 138 kV	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	168.2	213	42.7
ALGODONES SUBSTATION - ROYALTY 138 kV	FIRSTENERGY ALBRIGHT - SS3145EAGLE ROCK TAP 2 138 kV	157.2	252	43.8
MARACAIBO SUBSTATION - UNIVERSIITY 138 kV	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV Line	155.8	325	74.4

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HOLLY - BLUE RACER SUBSTATION 138 kV	BEDFORD - FIRSTENERGY ALBRIGHT 138 kV	145.5	182	25.5
HOLLY - CARSWELL 138 kV Line	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	141.2	193	25.5
MARACAIBO - CARSWELL 138 kV line	BEDFORD - FIRSTENERGY ALBRIGHT 138 kV	138.9	205	25.5
BLUE RACER - SIERRA PINE 138 kV	FRSTENERGY ALBRIGHT - BEDFORD SUBSTATION 138 kV L	137.0	184	25.5
KEIFER-JACKSON SUBSTATION - BEAVER DAM E 138 kV Line	TYRANENA - LG&E; CANAL 138 kV Line	130.8	251	9.0
LG&E; CANAL - TYRANENA 138 kV	KEIFER-ACKSON - BEAVER DAM EAST 138 kV	121.0	275	9.5
N SELMA - W OLEAN 138 kV	FIRSTENERGY ALBRIGHT - BEDFORD 138 kV	120.8	193	20.9
MARLOW - WEST MESQUITE 138 kV	FIRSTENERGY ALBRIGHT - SS3145EAGLE ROCK TAP 2 138 kV L	113.6	265	15.8
OLYMPIA BREWERY - ARVERNE 138 kV	ARVERNE SUBSTATION - TYRANENA 138 kV	110.4	173	18.1
BEAVER DAM E - COMOX TAP 138 kV line	BARRETT - WOLFE 500 kV	109.1	194	5.6
WEST LEBANON - OAKDALE POWER STA 138 kV	WISSOTA - NSP 138 kV Line	107.4	243	5.2
SCHERTZ - MET CTR 138 kV Line	FIRSTENERGY ALBRIGHT SUBSTATION - BEDFORD 138 kV	106.9	234	14.8
CLEVELAND - RICHARDSON 138 kV Line	CLEVELAND - RICHARDSON 138 kV Line	105.0	328	8.3
CLEVELAND - RICHARDSON 138 kV #2	RICHARDSON - CLEVELAND 138 kV	105.0	328	8.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
BEDFORD - BAXTER 138 kV	Rebuild line with higher-capacity conductor	\$8,467,000	100% (\$8,467,000)
FIRSTENERGY ALBRIGHT SUBSTATION - BEDFORD SUBSTATION 138 kV Line	Rebuild line with higher-capacity conductor	\$33,063,000	100% (\$33,063,000)
BAXTER - MARACAIBO 138 kV line	Rebuild line with higher-capacity conductor	\$9,689,000	100% (\$9,689,000)
EL PORTA - POSO MOUNTAIN 138 kV	Rebuild line with higher-capacity conductor	\$8,227,000	100% (\$8,227,000)
ZENITH SUBSTATION - FIRSTENERGY ALBRIGHT 138 kV	Rebuild line with higher-capacity conductor	\$3,225,000	98% (\$3,171,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SS3145EAGLE ROCK TAP 2 - ARVERNE 138 kV	Rebuild line with higher-capacity conductor	\$30,548,000	100% (\$30,548,000)
FIRSTENERGY ALBRIGHT SUBSTATION - SS3145EAGLE ROCK TAP 2 138 kV	Rebuild line with higher-capacity conductor	\$4,704,000	100% (\$4,704,000)
ZENITH - ROYALTY 138 kV	Rebuild line with higher-capacity conductor	\$7,390,000	98% (\$7,266,000)
UNIVERSITY SUBSTATION - EL PORTA 138 kV	Rebuild line with higher-capacity conductor	\$18,595,000	100% (\$18,595,000)
ROYALTY - ALGODONES 138 kV	Rebuild line with higher-capacity conductor	\$1,545,000	98% (\$1,519,000)
MARACAIBO SUBSTATION - UNIVERSITY 138 kV	Rebuild line with higher-capacity conductor	\$7,735,000	100% (\$7,735,000)
HOLLY - BLUE RACER SUBSTATION 138 kV	Rebuild line with higher-capacity conductor	\$4,069,000	100% (\$4,069,000)
HOLLY - CARSWELL 138 kV Line	Rebuild line with higher-capacity conductor	\$17,049,000	100% (\$17,049,000)
MARACAIBO - CARSWELL 138 kV line	Rebuild line with higher-capacity conductor	\$17,946,000	100% (\$17,946,000)
BLUE RACER - SERRA PINE 138 kV	Rebuild line with higher-capacity conductor	\$10,372,000	100% (\$10,372,000)
BEAVER DAM EAST - KEIFER-JACKSON 138 kV L	Rebuild line with higher-capacity conductor	\$4,321,000	60% (\$2,582,000)
LG&E; CANAL SUBSTATION - TYRANENA SUBSTATION 138 kV Line	Rebuild line with higher-capacity conductor	\$2,337,000	63% (\$1,478,000)
NORTH SELMA - W OLEAN 138 kV	Rebuild line with higher-capacity conductor	\$12,236,000	57% (\$6,981,000)
MARLOW - W MESQUITE 138 kV Line	Rebuild line with higher-capacity conductor	\$1,328,000	33% (\$437,000)
OLYMPIA BREWERY - ARVERNE 138 kV	Rebuild line with higher-capacity conductor	\$3,188,000	97% (\$3,078,000)
BEAVER DAM E - COMOX TAP 138 kV line	Reconductor line	\$47,887,000	49% (\$23,271,000)
W LEBANON - OAKDALE POWER STA 138 kV	Reconductor line	\$7,009,000	14% (\$966,000)
SCHERTZ - MET CTR 138 kV	Reconductor line	\$607,000	82% (\$497,000)
CLEVELAND - RICHARDSON 138 kV Line	Reconductor line	\$1,735,000	25% (\$427,000)
CLEVELAND - RICHARDSON 138 kV #2	Reconductor line	\$1,335,000	25% (\$328,000)
	Total allocated to project		\$222,465,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.